

1. Launch/configure an EC2 instance; create an S3 bucket with a lifecycle policy; write a least-privilege IAM policy for one role. Submit config + policy JSON.

Task 1: Launch/configure an EC2 instance

so basically go to AWS console & click on EC2. then u click Instances on the left side.

step 1: Launch Instances button

select an AMI -Ubuntu.

step 2: Pick Instance Type example use t2.micro

step 3: Instance Details

number of instances = 1. just use default VPC. make sure Auto-assign public IP is enabled

step 4: Leave storage default

step 5: Download Key Pair

click launch & it asks u to create/select key pair. create new one, name it my-ec2-keypair.

TASK 2 - Make S3 Bucket with Lifecycle Policy

Step 1:- Search S3.

Step 2:- Click Create bucket.

Step 3:- Enter unique bucket name.

Step 4:- Click Create bucket.

Step 5:- Open bucket -> Management.

Step 6:- Click Create lifecycle rule.

Step 7:- Give rule name.

Step 8:- Apply to all objects.

Step 9:- Expire current versions after 30 days.

Step 10:- Click Create rule.

TASK 3 - Create IAM Role & Least-Privilege Policy

Step 1:- Search IAM.

Step 2:- Open Policies.

Step 3:- Click Create policy.

Step 4:- Open JSON tab.

Step 5:- Paste JSON.

Step 6:- Replace YOUR_BUCKET_NAME.

Step 7:- Create policy.

Step 8:- Open Roles.

Step 9:- Attach the policy to the role.

THE LEAST-PRIVILEGE IAM POLICY JSON:

```
{ "Version": "2012-10-17",  
  "Statement": [ {  
    "Sid": "S3ReadOnlyAccess",  
    "Effect": "Allow",  
    "Action": [ "s3:GetObject", "s3:ListBucket", "s3:GetBucketVersioning" ],  
    "Resource": [ "arn:aws:s3:::my-bucket-12345", "arn:aws:s3:::my-bucket-12345/*" ]  
  } ]  
}
```